

## OBLUCHYE, Russian Federation

WMO: 317020

Lat: 49.0250N

Lon: 131.0493E

Elev: 267

StdP: 98.16

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |              |            |              | MCWS/PCWD to 99.6% DB |          | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|--------------|------------|--------------|-----------------------|----------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |              | 1%         |              |                       |          |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB         | WS         | MCDB         | MCWS                  | PCWD     |              |
| (a)<br>1      | (b)<br>-36.2 | (c)<br>-34.2 | (d)<br>-39.5                  | (e)<br>0.1 | (f)<br>-36.0 | (g)<br>-37.6 | (h)<br>0.1 | (i)<br>-34.1 | (j)<br>4.8            | (k)<br>-22.0 | (l)<br>4.3 | (m)<br>-23.4 | (n)<br>1.4            | (o)<br>0 | (p)<br>0.556 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>9.9             | (c)<br>29.4     | (d)<br>21.1 | (e)<br>27.6 | (f)<br>20.3 | (g)<br>26.1 | (h)<br>19.4 | (i)<br>22.6         | (j)<br>27.0 | (k)<br>21.6 | (l)<br>26.0 | (m)<br>20.6 | (n)<br>24.6 | (o)<br>2.0           | (p)<br>270 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 21.2                            | 16.4 | 24.9 | 20.1 | 15.3 | 24.0 | 19.2 | 14.4 | 23.0 | 68.1          | 27.0 | 64.5 | 26.2 | 61.1 | 24.6 | 29.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 5.3               | 4.5  | 4.0 | -39.0                      | 32.7 | 2.1                | 1.5 | -40.5                                              | 33.8 | -41.8      | 34.7 | -43.0      | 35.5 | -44.5      | 36.6 |
|                   |      |     | -39.1                      | 24.6 | 2.1                | 1.6 | -40.6                                              | 25.7 | -41.9      | 26.7 | -43.0      | 27.5 | -44.5      | 28.7 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | -0.2   | -24.1 | -19.2 | -9.2  | 2.8   | 11.0  | 17.0  | 20.4  | 18.4  | 11.8  | 2.0   | -11.7 | -22.5 |
|                                                                        | DBStd       | 16.49  | 4.52  | 5.33  | 6.48  | 4.76  | 3.80  | 3.40  | 2.44  | 2.95  | 3.88  | 4.93  | 7.12  | 5.02  |
|                                                                        | HDD10.0     | 4662   | 1056  | 818   | 596   | 222   | 33    | 1     | 0     | 1     | 26    | 252   | 652   | 1006  |
|                                                                        | HDD18.3     | 6896   | 1314  | 1052  | 854   | 467   | 228   | 64    | 8     | 36    | 198   | 507   | 902   | 1264  |
|                                                                        | CDD10.0     | 946    | 0     | 0     | 0     | 4     | 65    | 211   | 322   | 261   | 79    | 3     | 0     | 0     |
|                                                                        | CDD18.3     | 137    | 0     | 0     | 0     | 0     | 1     | 24    | 73    | 37    | 2     | 0     | 0     | 0     |
|                                                                        | CDH23.3     | 1162   | 0     | 0     | 0     | 7     | 53    | 284   | 532   | 268   | 18    | 0     | 0     | 0     |
|                                                                        | CDH26.7     | 263    | 0     | 0     | 0     | 1     | 8     | 78    | 134   | 41    | 1     | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 1.8    | 1.5   | 1.7   | 1.9   | 2.2   | 2.2   | 1.9   | 1.7   | 1.7   | 1.8   | 1.9   | 1.8   | 1.5   |
| (15) Precipitation                                                     | PrecAvg     | 789    | 9     | 10    | 25    | 44    | 84    | 111   | 161   | 156   | 89    | 52    | 30    | 18    |
|                                                                        | PrecMax     | 1065   | 36    | 35    | 71    | 132   | 187   | 204   | 242   | 253   | 169   | 102   | 91    | 64    |
|                                                                        | PrecMin     | 619    | 0     | 0     | 1     | 8     | 22    | 32    | 51    | 63    | 29    | 13    | 3     | 5     |
|                                                                        | PrecStd     | 111    | 8     | 9     | 18    | 26    | 38    | 49    | 46    | 51    | 34    | 22    | 19    | 13    |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | -8.4  | -1.7  | 9.0   | 23.5  | 27.6  | 32.0  | 32.1  | 29.6  | 25.7  | 19.4  | 6.8   | -6.2  |
|                                                                        |             | MCWB   | -9.9  | -4.5  | 2.8   | 10.6  | 14.4  | 20.5  | 21.8  | 22.3  | 18.4  | 11.0  | 2.4   | -7.4  |
|                                                                        | 2%          | DB     | -12.2 | -4.8  | 5.8   | 17.6  | 24.8  | 28.7  | 29.9  | 27.8  | 23.2  | 15.2  | 3.1   | -9.6  |
|                                                                        |             | MCWB   | -13.3 | -6.9  | 0.9   | 8.1   | 13.5  | 19.3  | 21.8  | 21.4  | 16.7  | 8.5   | -0.1  | -10.6 |
|                                                                        | 5%          | DB     | -14.5 | -7.4  | 3.2   | 14.2  | 22.1  | 26.5  | 28.0  | 26.2  | 21.2  | 12.3  | 0.6   | -11.6 |
|                                                                        |             | MCWB   | -15.3 | -9.1  | -0.7  | 6.4   | 12.5  | 18.4  | 21.4  | 20.4  | 15.4  | 6.6   | -1.1  | -12.4 |
|                                                                        | 10%         | DB     | -16.5 | -10.2 | 0.8   | 11.2  | 19.6  | 24.4  | 26.3  | 24.6  | 19.0  | 9.8   | -1.4  | -14.3 |
|                                                                        |             | MCWB   | -17.1 | -11.5 | -2.0  | 4.9   | 11.3  | 17.5  | 20.5  | 19.6  | 14.6  | 5.6   | -2.9  | -14.9 |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | -9.9  | -4.2  | 3.3   | 11.2  | 16.6  | 22.1  | 24.3  | 23.7  | 20.0  | 12.0  | 2.8   | -7.2  |
|                                                                        |             | MCDB   | -8.6  | -1.7  | 8.5   | 21.1  | 24.0  | 30.2  | 29.2  | 27.4  | 23.4  | 17.4  | 6.2   | -6.4  |
|                                                                        | 2%          | WB     | -13.3 | -6.8  | 1.2   | 8.8   | 14.9  | 20.4  | 22.8  | 22.4  | 18.0  | 9.5   | 0.5   | -10.5 |
|                                                                        |             | MCDB   | -12.3 | -4.9  | 4.9   | 16.4  | 21.6  | 26.3  | 27.4  | 26.2  | 21.2  | 13.5  | 2.3   | -9.6  |
|                                                                        | 5%          | WB     | -15.2 | -9.0  | -0.3  | 6.9   | 13.5  | 19.3  | 21.9  | 21.4  | 16.6  | 7.5   | -0.9  | -12.4 |
|                                                                        |             | MCDB   | -14.5 | -7.4  | 2.6   | 13.4  | 19.9  | 25.0  | 26.6  | 25.2  | 20.0  | 11.4  | 1.0   | -11.6 |
|                                                                        | 10%         | WB     | -17.1 | -11.5 | -1.8  | 5.1   | 12.2  | 18.2  | 21.1  | 20.2  | 15.2  | 5.9   | -3.0  | -14.9 |
|                                                                        |             | MCDB   | -16.5 | -10.2 | 0.9   | 10.5  | 18.6  | 23.5  | 25.2  | 23.5  | 18.1  | 9.3   | -1.4  | -14.2 |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 12.3  | 14.1  | 13.4  | 12.0  | 12.6  | 11.2  | 9.9   | 10.3  | 11.8  | 11.2  | 10.8  | 11.1  |
|                                                                        | 5% DB       | MCDBR  | 13.5  | 15.6  | 13.7  | 17.5  | 18.1  | 15.7  | 13.6  | 13.0  | 15.5  | 16.0  | 11.0  | 11.9  |
|                                                                        |             | MCWBR  | 12.5  | 13.5  | 9.1   | 9.5   | 8.7   | 7.3   | 6.7   | 6.9   | 8.8   | 9.3   | 8.2   | 11.1  |
|                                                                        | 5% WB       | MCDBR  | 13.5  | 15.2  | 12.3  | 16.1  | 14.9  | 13.2  | 11.6  | 11.0  | 12.6  | 14.1  | 10.2  | 11.8  |
|                                                                        |             | MCWBR  | 12.5  | 13.1  | 8.2   | 9.0   | 8.0   | 6.7   | 6.4   | 6.4   | 8.2   | 8.8   | 7.7   | 11.0  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.286 | 0.319 | 0.396 | 0.537 | 0.584 | 0.520 | 0.498 | 0.446 | 0.413 | 0.427 | 0.354 | 0.292 |
|                                                                        | taud        |        | 2.133 | 2.024 | 1.871 | 1.711 | 1.701 | 1.947 | 2.062 | 2.204 | 2.199 | 1.989 | 2.060 | 2.101 |
|                                                                        | Ebn at Noon |        | 746   | 801   | 787   | 710   | 698   | 755   | 766   | 791   | 776   | 660   | 634   | 670   |
|                                                                        | Edh at Noon |        | 82    | 118   | 165   | 216   | 227   | 179   | 157   | 131   | 118   | 123   | 90    | 75    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.82  | 3.06  | 4.51  | 4.83  | 5.15  | 5.50  | 5.13  | 4.68  | 3.80  | 2.54  | 1.70  | 1.42  |
|                                                                        | RadStd      |        | 0.05  | 0.07  | 0.28  | 0.51  | 0.49  | 0.74  | 0.49  | 0.34  | 0.38  | 0.19  | 0.12  | 0.07  |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (8 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page